Calculus item math and dependency audit

My role Math and item-quality review

I audited a 25-question calculus bank for math, answer logic, LaTeX, and variant dependencies. The review calls out six blocking errors and gives writers exact fixes.

ORIGINAL WORK FILE EXCERPT

5.3 Question Bank — Full Math & Dependency Audit (25 Questions)

Status legend: OK = no changes needed; Minor = cosmetic fix; Fix… = content issue; FIX (caps) = blocker (math or rendering).

Summary of blockers

• Q3 — interval typo: [0, 2/π] must be [0, 2π].

• Q7 — LaTeX for √(x+1) is broken: rendered as √x + 1 in choices A, B, D and solution.

• Q9 — Answer line has a broken bold/fraction wrapper ("**\frac{**4}{3}").

• Q11 — closing brace typo \right} instead of \right).

• Q20 — stem limit expression is malformed (exponent '2 · · 1/n', Σ swallowed by lim subscript); leftover author note 'ONLY 2 VARIANTS FOR THIS'.

• Q25 — extensive LaTeX never rendered as math (\\sqrt, \thinsp, \right} shown literally).

Recurring highlight issues

• Answer lines should not be highlighted. Violations: Q5, Q14, Q15, Q19, Q20, Q21, Q24 (and the dollar amount in Q24 only incidentally contains the variable digits).

• Same value highlighted in different colors for the same dependency: Q6 (2 as plateau height — yellow vs green), Q12 (the '2' of 2−4x